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EDUCATION

Riphah International University
BS, Software Engineering

Spring 2018

WORK EXPERIENCE

Trident Marketing
Tech Lead

Sep 2018 – Present

- Integrated mobile apps with external booking systems and third-party services to streamline business operations (REST, WebSockets, ElasticSearch).
- Architected scalable microservices and hybrid mobile solutions, leveraging Node.js, TypeScript, and MongoDB to support enterprise-level requirements. Designed system architecture patterns, API gateways, and authentication flows (JWT/OAuth) for secure, cloud-native applications.
- Designed and co-architected the TRA API to manage real-time data flow between multiple apps and external booking systems, leveraging REST API principles for scalability and efficiency. (Node.js, TypeScript).
- Designed the TRADB (MongoDB) schema for large-scale data handling and integrations.
- Optimized system performance and scalability by managing real-time synchronization across multiple Elastic indices, applying caching strategies, and tuning database queries.
- Enabled booking-to-sales conversion by updating household records, converting leads to members, and indexing data in the Manifest System.
- Built the Engage App using Angular and TypeScript, enabling agents to track customer data (e.g., demographics, call history) and manage bookings within the same platform, significantly streamlining campaign workflows and improving agent productivity.
- Engineered the Manifest System (Angular, Node.js) to track bookings, agent performance, commissions, and the overall sales lifecycle.
- Developed the DNQ System, a multi-part survey platform:
 - Backend uses a tree data structure for complex survey logic.
 - Frontend built with Ionic (Angular, Node.js).
- Contributed to the Lead Input Application (LIA) UI for uploading leads to S3 and splitting large PDFs (Angular, Node.js).
- Created various survey management platforms using Ionic.
- Implemented a message broker for high-volume calls, integrating with TRADB for faster processing.

- Built WebSocket-based solutions for real-time notifications, integrated with REST APIs to enhance scalability (e.g., member reservations).
- *Python*:
 - Deployed Google Cloud Vision to convert handwritten forms into digital data.
 - Used ChatGPT for data analysis, KPI derivation, and pre-filling forms.
 - Created a FastAPI-based rule engine for reservation eligibility using historical data.
 - Wrote behavior-driven tests using Behave.
 - Performed data cleaning with Pandas and built Jupyter notebooks for analysis.
- *Infrastructure & CI/CD*:
 - Optimized MongoDB cluster configurations, reducing operational costs by approximately saving \$12,000 annually while maintaining system performance and reliability. (MongoDB)
 - Led Kubernetes-based deployments and CI/CD pipeline improvements (Jenkins, GitHub Actions), incorporating Infrastructure as Code (IaC) practices to enhance scalability, resilience, and operational efficiency.
 - Maintained real-time sync with Elastic indices.
 - Created unit/behavior tests (Selenium, Chai, Mocha, Cypress) and integrated GitHub pipelines (JavaScript, Node.js).
 - Configured Jenkins pipelines for automated test reporting.

Code For Pakistan (Volunteer)

Mar 2021 – Present

Web Engineer/Flutter Developer

- Integrated a WhatsApp chatbot for patient surveys, enabling identification and random assignment of high-risk patients to volunteer doctors (React, Node.js, PostgreSQL).
- Built a Flutter-based mobile app for lab evaluations (LAARC-standard surveys) paired with a Vue.js dashboard for performance reporting (Flutter, Vue.js, Node.js).
- Collaborated with WHO and NIH as a Solutions Engineer to architect and implement secure healthcare applications (HCAI app) using Flutter, Vue.js, Node.js, and MongoDB. Applied system design principles, microservices architecture, and performance tuning while documenting architecture and workflows to improve healthcare protocols and post-surgery infection tracking. (Flutter, Vue, Node.js, MongoDB)
- Conducted training sessions for the KP Women Program, equipping participants with essential version control skills to enhance their collaborative development workflows.
- Facilitated the interview process in 2025 to hire mentors and candidates for the KP Women Program.

Jeddah Soft

Sep 2018 – Dec 2018

Mern Stack Developer (Remote Part Time)

- *Doctor App* (MySQL, React, Node.js):
 - Built a React + Bootstrap dashboard for managing doctor appointments.
 - Implemented Highcharts for admin reporting.
 - Maintained a session-based shopping cart tied to user authentication.
- *ShopAPI (Open Source)*:
 - See the ShopAPI GitHub repo (Express/Node.js + WebSockets).
 - Uploaded and indexed files/images via Cloudinary.

- Maintained session-based shopping cart logic.

xFlow Research Inc.

Oct 2016 – Feb 2017

Internee

- Developed a VR basketball game for HTC Vive.

Other Projects

Text to Speech Text to Speech

- Built real-time transcription microservices with WebSocket and REST API integrations, optimizing streaming and KPI extraction processes.
- Developed a microservice leveraging WebSockets to stream call recordings to Watson Speech-to-Text, enabling real-time transcription and automated KPI extraction.
- Designed the system to identify sensitive information, such as unintended requests for passcodes or restricted data, enhancing compliance and security.

Skills

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| • System Design & Architecture Patterns (Microservices, API Gateway, Event-driven Systems) | • Cross-functional Team Leadership & Stakeholder Communication |
| • JavaScript (Angular, React, Vue, Ionic) | • Performance Optimization & Scalability |
| • Node.js & TypeScript | • MongoDB, MySQL, MsSQL |
| • Python (FastAPI, Pandas, Jupyter) | • Docker, Kubernetes, CI/CD |
| • GraphQL | • Jenkins, GitHub Actions |
| • Flutter, ionic | • AWS (S3, Elastic) AWS Certified |
| • Selenium, Cypress, Protractor, Mocha, Chai | • Message Brokers, WebSockets |
| • BDD (Cucumber, Behave) | • Git (GitHub, GitLab) |
| | • Google Cloud Vision, ChatGPT |

School Projects

Honeypot Analysis and Visualization Tool (FYP)

- Analyzed data from honeypots (Dionaea, Suricata, Kippo).
- Visualized data intensity on geographical maps.
- Centralized data in MongoDB for increased performance.

Encrypted Document Vault with Sharing Control (Distributed Secure Platform)

- Engineered a secure, distributed document vault platform for sharing sensitive files with full owner control, utilizing Go (Golang) for high-performance backend services and Angular/JavaScript for a responsive frontend.

- Designed a multi-instance distributed architecture to support horizontal scaling, ensuring system-wide state consistency (e.g., document deletion/access revocation) by using RabbitMQ (AMQP) for reliable event dispatching across all instances.
- Implemented a real-time “Eye Tracking System” via WebSockets to instantly notify document owners of current viewers, tracking access by IP address and browser fingerprint for traceability.
- Enforced comprehensive security, including AES-256 file encryption with user-specific keys, configurable time-limited/burn-once links, and instant system-wide access revocation.
- Integrated anti-capture measures including watermarking and client-side hardening to restrict copying, downloading, and developer tool access within the document view.
- Utilized Docker and Docker Compose for containerization and deployment, structuring the backend with a clean repository pattern using Go generics for testability and separation of concerns.

Achievements/Interests

- Ten Community Alumni Team Lead
- Problem Solving